

## Location Privacy

In subject area: [Computer Science](#)

Location privacy refers to individuals' ability to move in public spaces without their locations being disclosed, ensuring that their spatial characteristics and identity remain unlinked. It involves protecting one's location information from being associated with their actual identity.

**AI generated definition based on:** [Vehicular Communications, 2022](#)

How useful is this definition?   

### On this page

#### Common questions

AI-generated

What are the privacy concerns related to Internet of Things device privacy? 

What is the Secure Cooperative Intersection Driver and how does it protect the privacy of drivers and passengers in Vehicular Cyber-Physical Systems? 

How can location privacy be enhanced using the perturbation mechanism of Differential Privacy? 

What are obfuscation techniques? 

What is location privacy? 

#### Chapters and Articles

You might find these chapters and articles relevant to this topic.

Review article

### Impacts of location-privacy preserving schemes on vehicular applications

[2022, Vehicular Communications](#)

Abdueli Paulo Mdee, ... Dongkyun Kim

#### 2.2.1 Overview of location privacy

A concise definition of location privacy is important for understanding, protecting, and measuring location privacy. Thus,



normal circumstances, to move in public spaces without worrying about their locations being disclosed. It implies that individuals expect their locations to be protected when moving in public spaces (where there are different actors, benign and malicious). Furthermore, some circumstances require individuals' locations to be recorded and disclosed; therefore, complete [anonymity](#) is not desirable.

In [9] [28], location privacy is defined as the extent to which an individual's spatial characteristics and identity cannot be linked together. This means that one of the ways to achieve location privacy is to disconnect the [location information](#) from its identity altogether. By doing so, it won't be easy to associate it with the actual individual. Other researchers claim that location privacy should ensure that no one knows precisely where the sender of information

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Chapter

## Sensor security and location privacy

[2017, Mobile Sensors and Context-Aware Computing](#)

Manish J. Gajjar

### Location Privacy

Location privacy means an ability to prevent unauthorized agents/users from learning the device's current or past location.

LBSs are services that use sensors to identify the device location and other user or device context information (such as the day, time, or environment). LBSs are a requirement today for [mobile computing](#) devices, because they offer device users information and services that are relevant to the user/device location.

The sensors or [sensor nodes](#) (in a network of sensors that provides location and context information) and their physical location is useful in [authentication](#), identifying geographic path of user, and other location-based applications. The geographic path of the user can provide information on the user's preference of places to visit, amount of time spent at those places, information about meetings or gatherings attended there, and so on. Since such location-based information is important and valuable it requires to be treated with caution, otherwise it could be obtained by an attacker resulting in unforeseen consequences. While using location-based [social media services](#) it is important to make decisions on which type of information the user desires to disclose and to whom. A user may not want to provide their [location information](#) to even their closest

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Chapter

## Privacy on the Internet

2013, [Computer and Information Security Handbook \(Second Edition\)](#)

Marco Cremonini, ... Claudio Agostino Ardagna

### Privacy in Mobile Environments

The concept of *location privacy* can be defined as the right of individuals to decide how, when, and for which purposes their [location information](#) could be released to other parties. The lack of location privacy protection could be exploited by adversaries to perform various attacks [46]:

- *Unsolicited advertising*, when the location of a user could be exploited, without her consent, to provide advertisements of products and services available nearby the user position.
- *Physical attacks or harassment*, when the location of a user could allow criminals to carry out physical assaults on specific individuals.
- *User profiling and tracking*, when the location of a user could be used to infer other [sensitive information](#), such as state of health, personal habits, or professional duties, by correlating visited places or paths.
- *Political, religious, sexual persecution and discrimination*, when the location of a user could be used to reduce the freedom of individuals, and mobile technologies are used to identify and persecute opponents.
- [Denial of service](#), when the location of a user could motivate an access denial to services under some circumstances.

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## A survey on location privacy protection in Wireless Sensor Networks

2019, *Journal of Network and Computer Applications*

Jinfang Jiang, ... Mohsen Guizani

### 2.1 Basic definitions

The expression of “privacy” is from the social network, where it is first defined as “the right to be left alone”. Then, a new conception of “information privacy” and “location privacy” are emerging. The “information privacy” concerns the personal privacy information, while “location privacy” means the possibility of approaching others' surroundings or private space (Smith et al., 2010). In this survey, we focus on the research of sensor nodes' location privacy problem. Other basic terminologies used in the location privacy preserving have been listed as follows.

**Base station:** It is also named a sink node, which is one of the most important [sensor nodes](#) in the [WSNs](#). As a manager node in WSNs, the base station or the sink node is responsible for collecting and aggregating all the packets or information in the WSNs.

**Source node:** It is a [sensor node](#) which senses information about environment or data about monitoring targets, and then tries to send all the sensed packets to the sink by multi-hop [routing paths](#).

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Review article

## Impacts of location-privacy preserving schemes on vehicular applications

2022, *Vehicular Communications*



## 2.2 Location privacy in vehicular networks

This subsection defines location privacy. In addition, it discusses privacy requirements, [adversary models](#), and location privacy attacks.

### 2.2.1 Overview of location privacy

A concise definition of location privacy is important for understanding, protecting, and measuring location privacy. Thus, different researchers have tried to define location privacy. In [27], the authors define location privacy as individuals' ability, under normal circumstances, to move in public spaces without worrying about their locations being disclosed. It implies that individuals expect their locations to be protected when moving in public spaces (where there are different actors, benign and malicious). Furthermore, some circumstances require individuals' locations to be recorded and disclosed; therefore, complete [anonymity](#) is not desirable.

In [9] [28], location privacy is defined as the extent to which an individual's spatial characteristics and identity cannot be linked together. This means that one of the ways to achieve location privacy is to disconnect the [location information](#) from its identity altogether. By doing so, it won't be easy to associate it with the actual individual. Other researchers claim that location privacy should ensure that no one knows precisely where the sender of information

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Review article

## Blockchain technology as a Fog computing security and privacy solution: An overview

2022, [Computer Communications](#)

Yehia Ibrahim Alzoubi, ... Hasan Kahtan

### 3.5.3 Location privacy

Location privacy is the third aspect of [FC](#) privacy that should be paid attention to. Only authorized nodes need to know the location of nodes delivering or receiving data. Patwary et al. [119] proposed a secure decentralized location-based device-to-device (D2D) [authentication](#) model in which Fog nodes can mutually authenticate each other at the Fog layer by using [BC](#). Through experimental



Kong et al.[63] reported that, in vehicular Fog, because the sensory data obtained is strongly location-correlated, it may reveal the street level position. Information, as well as maintaining vehicle location privacy implies that the content of each sensory input should be protected. Moreover, because the sensory data content space is restricted, the exhaustive approach may still be used to extract the location privacy of a data report from the matching signature by repeatedly attempting alternative sensory data values. As a result, location privacy must be ensured. Accordingly, the authors developed a privacy-preserving scheme using permissioned BC to verify sensory [data collecting](#) and sharing. The author reported that the verifiable computation of the average and variance of vehicular sensory data may be performed by utilizing the [cryptosystem](#) and an identity-based signature technique. In addition, the permissioned

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Review article

## Privacy preservation for V2G networks in smart grid: A survey

2016, [Computer Communications](#)

Wenlin Han, Yang Xiao

### 4.6 Location privacy

An EV can move from place to place. If an EV's [location information](#) is exposed to an adversary, the adversary could obtain its movement information and predict its next stop. If location privacy is not well protected, it could be used for a malicious purpose, such as hijacking.

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Chapter

## Spatial Privacy Challenges in Social Networks

2016, [Big Data](#)



## 11.5 Spatial Privacy Case Studies

As described previously, we wish to explore a range of questions related to location privacy:

- How can a location lose privacy?
- How easily can we reveal location privacy?
- What are the potential consequences of a loss of location privacy?

It is obvious that location privacy can be easily given away as shown in Fig. 1. If location-based services are used, then the privacy of an individual's exact location is immediately available. Users (tweeters) may be fully conscious of this and happy to release such data; however, other users may simply be unaware of exactly how much data they are releasing. The question arises, how are those users actually identified in the first place? Location privacy is one common route to discovering such individuals.

Given the general definition of privacy as the right of both being free from secret surveillance and regulating rules of accessing personal/organizational identity [32], a possible definition of a location losing privacy might be "actively revealing a location's identity." The obvious identity of a particular location includes the location name, the location's usage, for example, is the location used for [business activities](#) or [recreational activities](#), and the events that

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Review article

## Privacy preservation for V2G networks in smart grid: A survey

2016, [Computer Communications](#)

Wenlin Han, Yang Xiao

## 6.5 Identity and location privacy

The techniques employed to address the problems of identity privacy and location privacy overlap with the techniques of [anonymous authentication](#), charging unlinkability, and privacy-preserving billing and payment, and as do the references. The reason is that these papers [34,35,41–45,49–52,62,63] all hide an EV's real



solved by hiding the IDs in the papers [34,35,41–45,49–52,62,63].

The paper [31] addressed location privacy problem by hiding an EV's location directly. However, it cannot address identity privacy problem at the same time.

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Mini review

## A survey of broadcast authentication schemes for wireless networks

2015, *Ad Hoc Networks*

Kanika Grover, Alvin Lim

### 6.3 Location privacy and data confidentiality in VANETs

Another important new component is the privacy of the user location which is very relevant in [vehicular networks](#). When [cloud computing](#) is used in vehicular networks, some users would want to maintain their location privacy. Hence, the use of IDs and certificates will compromise the confidentiality and privacy of the sender and reveal sender's [personal data](#). Therefore, [privacy techniques](#) need to be implemented in [VANETs](#), with the condition that they will not add to the delay in [authentication](#) of [warning messages](#).

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### Related terms:

[Authentication](#), [Internet of Things](#),

[Privacy Preserving](#), [Anonymity](#),

[Differential Privacy](#), [Location-Based Service](#),

[Location Information](#), [Privacy Protection](#),

[Blockchain](#), [vehicular ad-hoc network](#).

### Recommended publications

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|                                                                                                                                                                                             | <b>Journal of Network and Computer Applications</b> |
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